Probability
Lecture 6: Random Vectors (Part 2)

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Content

1. Sum of independent random variables
2. Covariance matrix
3. Conditional Distribution

Previous lecture

- Recall that:

+ Random vector

$$X = (X_1, \ldots, X_n) : \ \Omega \to \mathbb{R}^n$$
$$\omega \mapsto X(\omega) = (X_1(\omega, \ldots, X_n(\omega)).$$

+ Discrete vs Continuous.

+ Two random variables X and Y are **independent** if for any measurable subsets $A, B \subset \mathbb{R}$

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

+ The **Covariance** of X and Y is defined as

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X].\mathbb{E}[Y].$$

.

Linearity of Expectation

- $\mathbb{E}[X+Y] = E[X] + \mathbb{E}[Y]$.
- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X,Y) + \mathrm{Var}(Y)$.
- If X and Y are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y];$$

and

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Sum of two independent Poisson variables

Theorem

Consider two independent random variables with Poisson distribution: $X_1 \sim Poi(\lambda_1)$ and $X_2 \sim Poi(\lambda_2)$. Then

$$X_1 + X_2 \sim Poi(\lambda_1 + \lambda_2).$$

For any $k \in \mathbb{N}$,

$$\mathbb{P}(X_1 + X_2 = k) = \sum_{l=0}^{k} \mathbb{P}(X_1 = l, X_2 = k - l) = \sum_{l=0}^{k} \mathbb{P}(X_1 = l)\mathbb{P}(X_2 = k - l)$$

$$= \sum_{l=0}^{k} \frac{\lambda_1^l e^{-\lambda_1}}{l!} \frac{\lambda_2^{k-l} e^{-\lambda_2}}{(k-l)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{l=0}^{k} \frac{k!}{l!(k-l)!} \lambda_1^l \lambda_2^{k-l}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1 + \lambda_2)^k}{k!}$$

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$$\begin{aligned}
\mathbb{P}(X_1 + X_2 = k) &= \sum_{l=0}^{k} \mathbb{P}(X_1 = l, X_2 = k - l) = \sum_{l=0}^{k} \mathbb{P}(X_1 = l)\mathbb{P}(X_2 = k - l) \\
&= \sum_{l=0}^{k} \frac{\lambda_1^l e^{-\lambda_1}}{l!} \frac{\lambda_2^{k-l} e^{-\lambda_2}}{(k-l)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{l=0}^{k} \frac{k!}{l!(k-l)!} \lambda_1^l \lambda_2^{k-l} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1 + \lambda_2)^k}{k!}
\end{aligned}$$

Sum of two independent continuous variables

Theorem

Consider two independent continuous random variables with density functions $f_X(x)$ and $f_Y(y)$. Let $Z = X + Y$. Then the density function (PDF) of Z is given by

$$f_Z(z) = (f_X * f_Y)(z) = \int_{\mathbb{R}} f_X(z-y) f_Y(y) dy = \int_{\mathbb{R}} f_X(x) f_Y(z-x) dx.$$

$$\mathbb{P}(X+Y \leq z) = \int_{\mathbb{R}} dy \left(\int_{-\infty}^{z-y} f_{X,Y}(x,y) dx \right) = \int_{\mathbb{R}} dy \left(\int_{-\infty}^{z-y} f_X(x) f_Y(y) dx \right)$$

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Examples

- Let X and Y be independent, and let

$$f_X(x) = \begin{cases} xe^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \text{and } f_Y(y) = \begin{cases} ye^{-y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Find the PDF of $Z = X + Y$.

- We have $f_Z(z) = \int_{\mathbb{R}} f_X(z-y) f_Y(y) dy$.

+ Then for $z < 0$, $f_Z(z) = 0$. And for $z \geq 0$

$$f_Z(z) = \int_0^z (z-y) e^{-(z-y)} y e^{-y} dy = e^{-z} \int_0^z (z-y) y dy = \frac{z^3}{6} e^{-z}.$$

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Examples

Theorem

Let X_1 and X_2 be two indendenpent Gaussian random variables such that

$$X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2) \text{ and } X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2).$$

Then

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Example of product of two random variables

- Let X and Y be independent, and let

$$f_X(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & \textit{otherwise} \end{cases} \quad \text{and } f_Y(y) = \begin{cases} 1, & 0 \le y \le 1 \\ 0, & \textit{otherwise} \end{cases}$$

Find the PDF of $Z = XY$.

- We have $F_Z(z) = \mathbb{P}(XY \le z)$. Then $F_Z(z) = 0$ for $z < 0$. For $z > 0$,

$$F_Z(z) = \mathbb{P}(XY \le z) = \int_0^1 dy \left(\int_0^{z/y} f_X(x) f_Y(y) dx \right)$$

+ We just consider $z \le 1$ since for $z > 1$, $F_Z(z) = 1$.

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \int_z^1 \frac{1}{y} f_X(z/y) f_Y(y) dy = 2(1-z).$$

- $f_Z(z) = \begin{cases} 2(1-z), & 0 \le z \le 1 \\ 0, & \textit{otherwise} \end{cases}$.

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1 Sum of independent random variables

2 Covariance matrix

3 Conditional Distribution

Covariance matrix

Definition

Given a random vector $X = (X_1, \ldots, X_n)$.

- Its expectation is

$$\mathbb{E}[X] = \mu = [\mathbb{E}[X_1], \mathbb{E}[X_2], \ldots, \mathbb{E}[X_n]].$$

- Its **covariance matrix** is given by $\Sigma = [(X - \mu)(X - \mu)^T]$

$$\Sigma = \mathrm{Cov}(X) = \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \ldots & \mathrm{Cov}(X_1, X_n) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \ldots & \mathrm{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \mathrm{Cov}(X_n, X_2) & \ldots & \mathrm{Var}(X_n) \end{pmatrix}.$$

Multidimensional Gaussian random vectors

Definition

A d-dimensional joint Gaussian random vector has the joint density function

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^d \det(\Sigma)}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right),$$

where μ: expectation vector and Σ: covariance matrix.

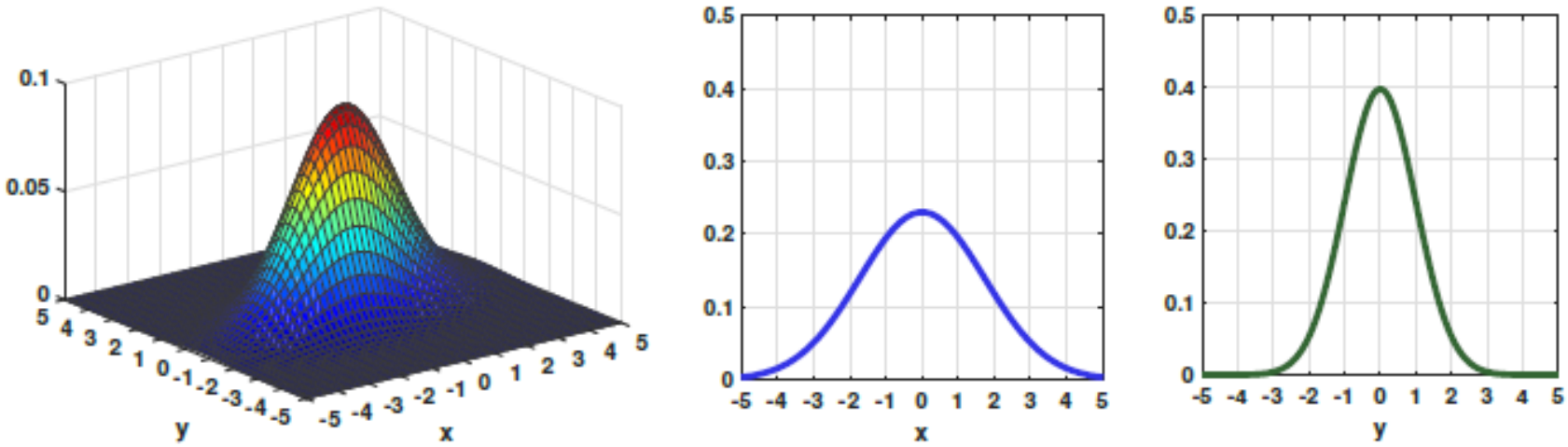
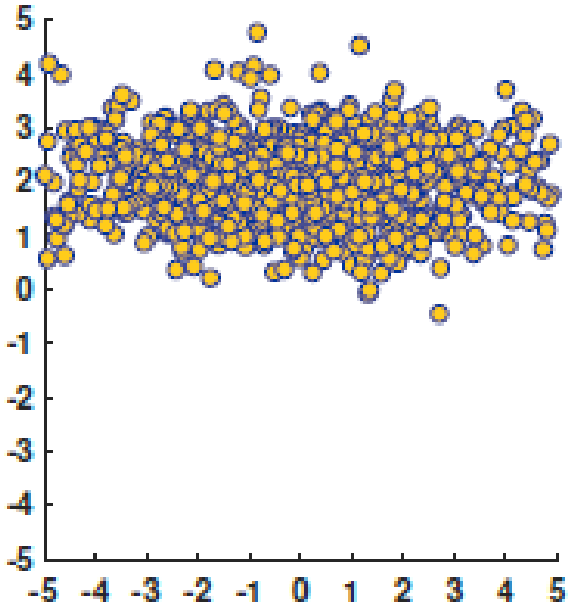
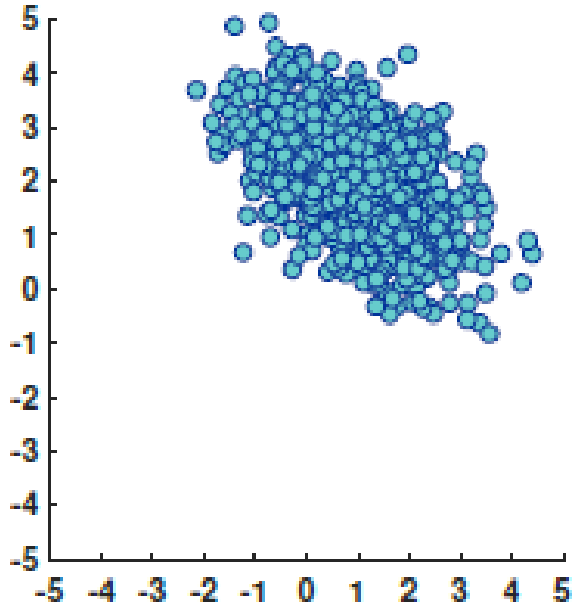


Figure: PDF of a two-dimensional Gaussian vector

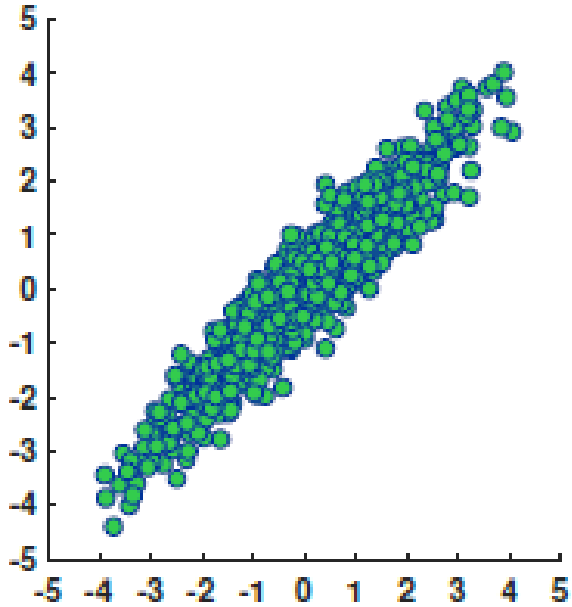
Example



$$(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



$$(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$



$$(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 & 1.9 \\ 1.9 & 2 \end{bmatrix}$$

Figure: Visualization of 2D Gaussians with different means and covariances.

Transformation of Gaussian vectors

Theorem

If $X \sim \mathcal{N}(\mu, \Sigma_X)$, *then for any* $A \in \mathbb{R}^{n \times n}$ *and* $b \in \mathbb{R}^n$,

$$Y = A.X + b \sim \mathcal{N}(A\mu + b, A\Sigma_X A^T).$$

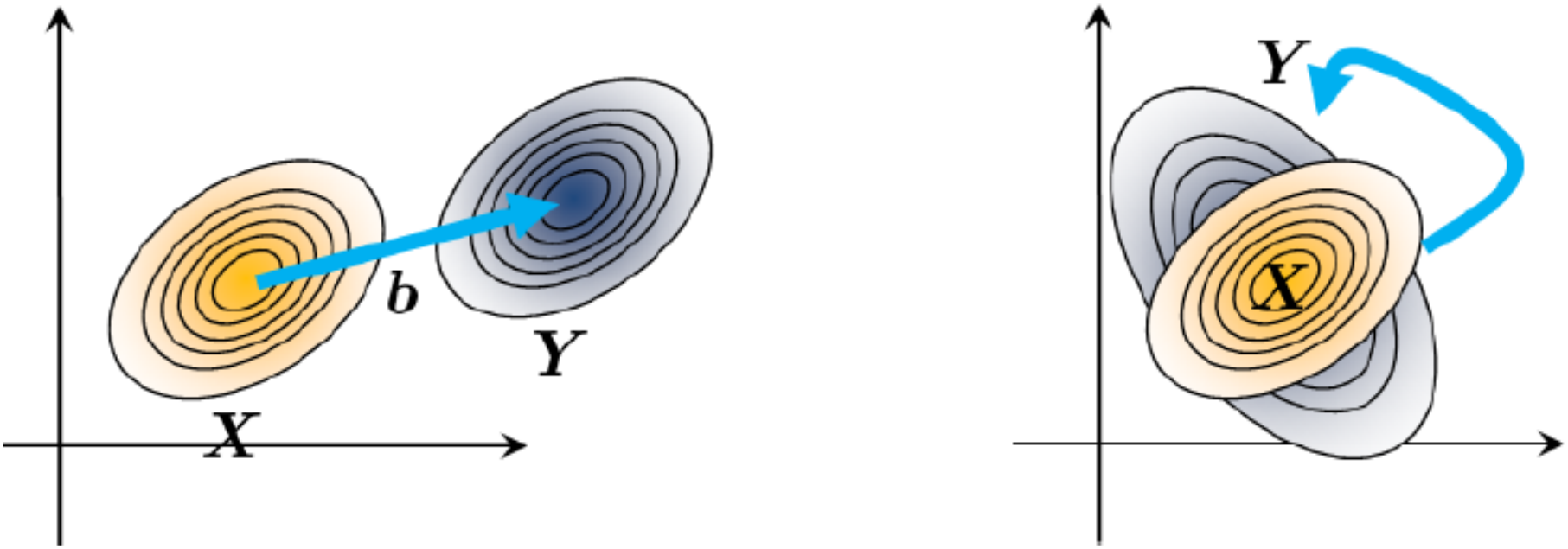


Figure: Transforming a Gaussian.

Recall on Linear Algebra

Definition

Given a square matrix $A \in R^{n \times n}$, the vector $u \in \mathbb{R}^n$, $u \neq 0$ is called the **eigenvector** of A if

$$Au = \lambda u,$$

for some $\lambda \in \mathbb{R}$. The scalarλ is called the **eigenvalue** associated with u. To find the eigenvalues of A, we solve

$$\det(A - \lambda I_n) = 0.$$

Theorem

If A is a symmetric square matrix, then all the eigenvalues of A are real, and there exist an orthogonal matrix U ($UU^T = I_n$) and a diagonal matrix Λ such that

$$A = U\Lambda U^T.$$

The row (or column) vectors of U can form a basis of $\mathbb{R}^n$.

Transformation of Gaussian vectors

Theorem

- Let $X \sim \mathcal{N}(0, I_n)$. Given a vector μ and a semi-positive definite (covariance) matrx Σ with eigendecomposition $\Sigma = U\Lambda U^T$. Then

$$Y = \Sigma^{1/2} X + \mu \sim \mathcal{N}(\mu, \Sigma),$$

where $\Sigma^{1/2} = U\Lambda^{1/2}U^T$.

- Conversely, let $Y = \Sigma^{1/2} X + \mu \sim \mathcal{N}(\mu, \Sigma)$. Then

$$X = \Sigma^{-1/2}(Y - \mu) \sim \mathcal{N}(0, I_n).$$

Gaussian whitening process

- **Gaussian whitening process**:
  + Step 1: Generating samples $\{x_1, \ldots, x_k\}$ from standard Gaussian $\mathcal{N}(0, I_n)$.
  + Step 2: Define $y_i = \Sigma^{1/2} x_i + \mu$ to simulate the Gaussian distribution $\mathcal{N}(\mu, \Sigma)$.

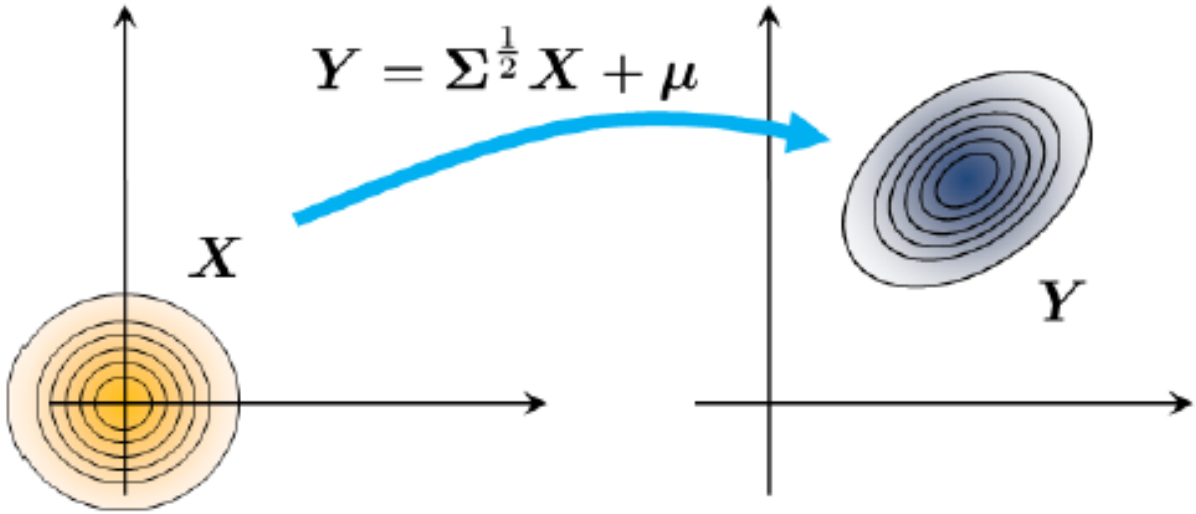


Figure: Generating an arbitrary Gaussian from Gaussian$(0, I_n)$.

Simulation

Simulate $Y \sim \mathcal{N}(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 3 & -0.5 \\ -0.5 & 1 \end{pmatrix}$$

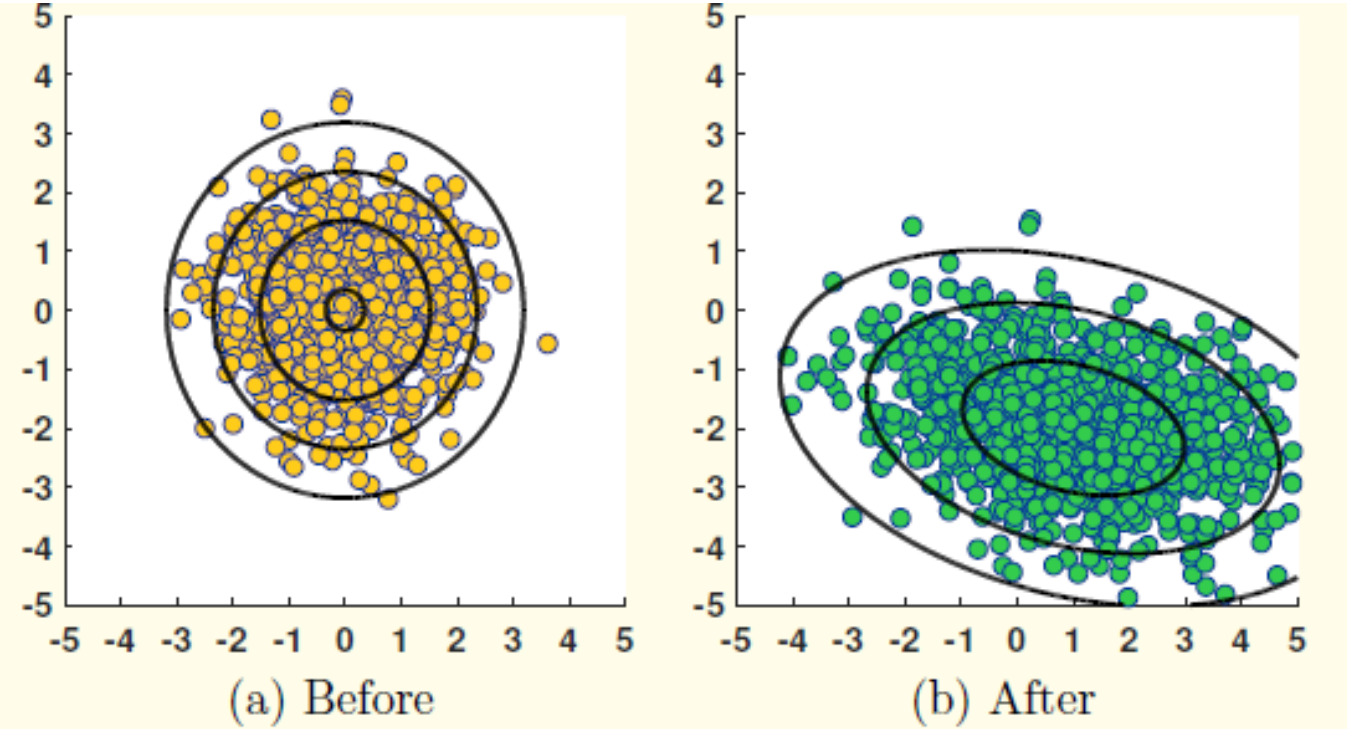


Figure: Emxample.

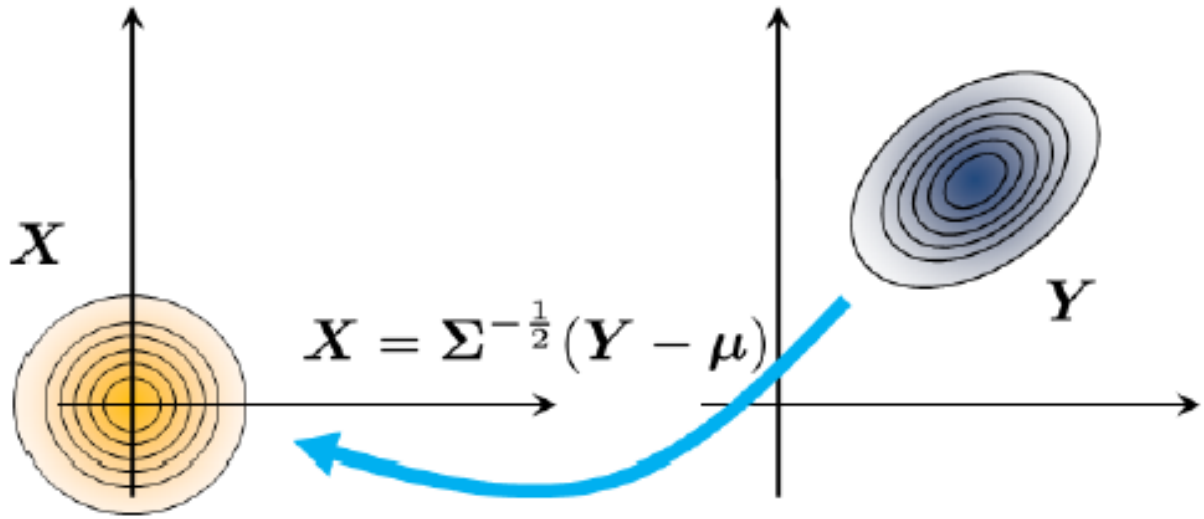


Figure: Converting an arbitrary Gaussian back to Gaussian$(0, I_n)$.

Principal Component Analysis (PCA)

- For any data point $x_i \in \mathbb{R}^n$, approximate with respect to a collection $\{v_1, \ldots, v_p\}$

$$x_i \approx \alpha_{i,1} v_1 + \ldots + \alpha_{i,p} v_p.$$

- Then, define a new representation

$$x_i \approx (\alpha_{i,1}, \ldots, \alpha_{i,p}).$$

$p < n$: Dimensionality Reduction!

$\rightarrow$ How to choose the right collection???

$$\operatorname*{argmin}_{\|v\|=1} \mathbb{E}[\|X - \alpha v\|^2] = \operatorname*{argmax}_{\|v\|=1} v^T \Sigma v.$$

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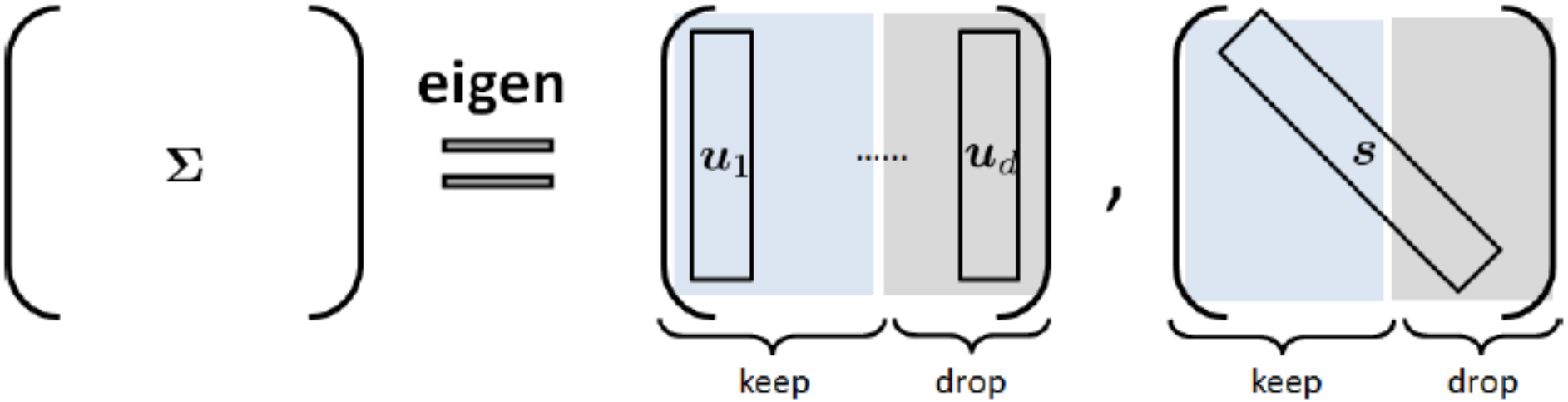
Principal Component Analysis (PCA)

Theorem

Let Sigma be with eigendecomposition $\Sigma = U\Lambda U^T$.. *Then the optimization*

$$\hat{v} = \underset{\|v\|=1}{argmax}\, v^T \Sigma v$$

has a solution $\hat{v} = u_1$, *the first column of the eigenvector matrix U*
Note that: we arrange $\lambda_1 \geq \lambda_2 \geq .....$



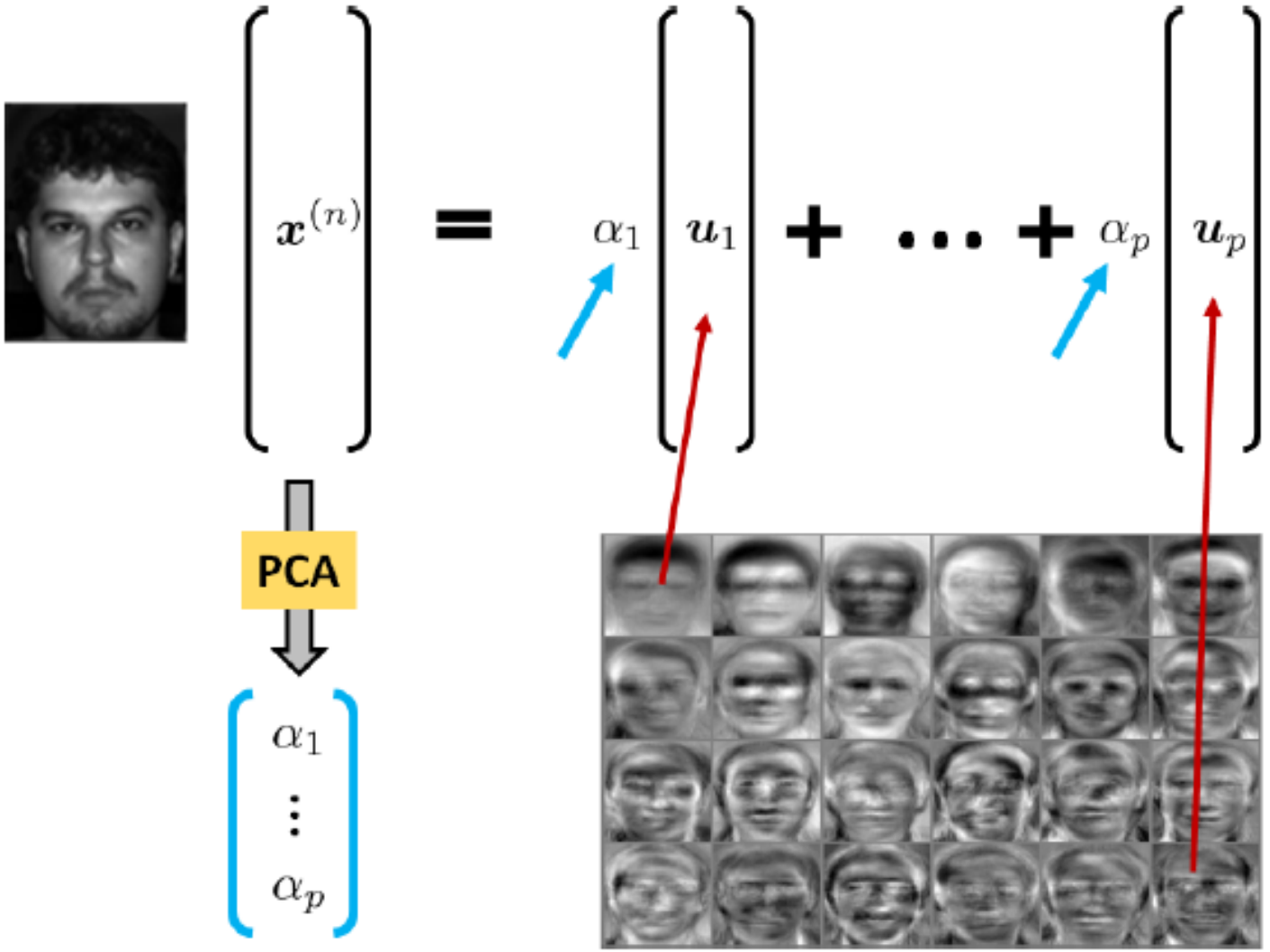
Computer Vision

-http://vision.ucsd.edu/ leekc/ExtYaleDatabase/ExtYaleB.html

$$n = 32'256, \quad N = 16'128.$$



$p = 100$



1 Sum of independent random variables

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Conditional Distribution

- Recall: for two events A and B (with $\mathbb{P}(B) > 0$), the conditional probability

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Definition

Let X and Y be two discrete random variables. The conditional probability mass function PMF of X given $Y = y$ is

$$\mathbb{P}_{X|Y}(x|y) = \frac{\mathbb{P}_{X,Y}(x,y)}{\mathbb{P}_Y(y)} = \mathbb{P}(X = x|Y = y).$$

Idea: we fix $Y = y$, X still obtains the values X_i but with different probability mass function

Example

- Suppose there are two coins. Let

X = the sum of the values of two coins,

Y = the value of the first coin.

Find the conditional PMF of X given $Y = 1$.

X	0	1	2
$\mathbb{P}_{X,Y}(x\|1)$	0	2/3	1/3

Example

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Continuous Conditional Distribution

Definition

Let X and Y be two continuous random variables. The conditional probability density function PDF of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_{X,Y}(x,y)}{\int_{\mathbb{R}} f_{X,Y}(x,y)dx}.$$

Let X and Y be two continuous random variables with a joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq y \leq x < \infty \\ 0, & otherwise \end{cases}.$$

Find the conditional PDFs $fX|Y(x,y)$ and $f_{Y|X}(y|x)$.

$f_X(x) = \int_0^x 2e^{-(x+y)}dy = 2e^{-x}(1-e^{-x}), f_Y(y) = \int_y^\infty 2e^{-(x+y)}dy = 2e^{-2y}$.

$$f_{X|Y}(x|y) = \frac{2e^{-(x+y)}}{2e^{-2y}} = e^{-x+y}, \quad f_{Y|X}(y|x) = \frac{e^{-y}}{1-e^{-x}}.$$

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Definition

Let X and Y be two continuous random variables. The conditional probability density function PDF of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_{X,Y}(x,y)}{\int_{\mathbb{R}} f_{X,Y}(x,y)dx}.$$

Let X and Y be two continuous random variables with a joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq y \leq x < \infty \\ 0, & otherwise \end{cases}.$$

Find the conditional PDFs $fX|Y(x,y)$ and $f_{Y|X}(y|x)$.
$f_X(x) = \int_0^x 2e^{-(x+y)}dy = 2e^{-x}(1-e^{-x}), f_Y(y) = \int_y^\infty 2e^{-(x+y)}dy = 2e^{-2y}$.

$$f_{X|Y}(x|y) = \frac{2e^{-(x+y)}}{2e^{-2y}} = e^{-x+y}, \quad f_{Y|X}(y|x) = \frac{e^{-y}}{1-e^{-x}}.$$

Example

Let $X = \pm 1$ be a random bit with equal probability. Suppose that X is transmitted over a noisy channel so that the observed signal is

$$Y = X + N$$

where the noise $N \sim \mathcal{N}(0, 1)$ is independent of the signal X. Find the probabilities $\mathbb{P}[X = +1 | Y > 0]$ and $\mathbb{P}[X = -1 | Y > 0]$.

- We have

$$\mathbb{P}(Y > 0 | X = 1) = \mathbb{P}(\mathcal{N}(1, 1) > 0) = 1 - \Phi(-1) = 0.8413.$$

$$\mathbb{P}(Y > 0 | X = -1) = \mathbb{P}(\mathcal{N}(-1, 1) > 0) = 1 - \Phi(1) = 0.1587.$$

$$\mathbb{P}(Y > 0) = 1/2.$$

$$\mathbb{P}[X = +1 | Y > 0] = 0.8413; \quad \mathbb{P}[X = -1 | Y > 0] = 0.1587.$$

Observe that output $Y > 0$, what is our opinion on the input X?
$\rightarrow$ Bayesian classification, Bayesian statistics.

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Thank you!